PRODUCT USER MANUAL

For OMI_CLIMATE_SI_ARCTIC_transport

Issue: 3.2

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2021/01/10	all	Creation of the document	R. P. Raj	V.S. Lien
2.0	2021/09/10	all	Update of the document	V.S. Lien	R. P. Raj
3.0	2022/08/15	all	Change from TOPAZ4a to TOPAZ4b; change in naming convention	R. P. Raj	V.S. Lien
3.1	2025/05/25	all	Update of the document structure	R. P. Raj; V.S. Lien	L. Bertino

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GLOSSARY AND ABBREVIATIONS

OMI	Ocean Monitoring Indicator
NetCDF	Network Common Data Form
PUM	Product User Manual
QUID	Quality Information Document

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is the user manual for the Copernicus Marine Service Ocean Monitoring Indicator (OMI) OMI_CLIMATE_SI_ARCTIC_transport.

The time series are computed using the TOPAZ4b reanalysis from the Arctic Marine Forecasting Centre (product name ARCTIC_MULTIYEAR_PHY_002_003).

The product is published on the Marine Data Store after automatic and delayed-mode quality controls. The product is available online and disseminated through the Marine Data Store. Files downloaded are in NetCDF format and follow CF-1.7 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

Product OMI_CLIMATE_SI_ARCTIC_transport is described in the Quality Information Document (QUID) CMEMS-ARC-QUID-CLIMATE-SI-ARCTIC-transport-v3.1

Information on operational issues on products and services can be found on our **User Notification Service**. If you have any questions, please **contact us**.

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	OMI_CLIMATE_SI_ARCTIC_transport
Geographical coverage	Fram Strait and Barents Sea (76°N - 80°N; 20°W - 60°E)
Variables	SI_AT_FS_Reanalysis SI_AT_BSE_Reanalysis SI_AT_BSN_Reanalysis SI_VT_FS_Reanalysis SI_VT_BSE_Reanalysis SI_VT_BSN_Reanalysis
Available time series	1991-01-15 to 2024-12-15
Temporal resolution	Monthly means
Horizontal resolution	N/A
Integration depth (for example)	Surface
Format	NetCDF4

b) Production System Description

This product provides the modelled time series of monthly averaged sea-ice area and volume transports across the Fram Strait, northern and eastern Barents Sea derived from the Arctic reanalysis product ARCTIC_MULTIYEAR_PHY_002_003.

c) Dataset details

OMI_CLIMATE_SI_ARCTIC_transport contains the monthly averaged sea-ice transport between the Polar Basin and the Nordic and Barents seas
SI_AT_FS_Reanalysis [km^2 month^{-1}] Sea ice area transport across the Fram Strait, from reanalysis sea_ice_transport_across_line **SI_VT_FS_Reanalysis [km^3 month^{-1}]** Sea ice volume transport across the Fram Strait, from reanalysis sea_ice_transport_across_line **SI_AT_BSE_Reanalysis [km^2 month^{-1}]** Sea ice area transport across the eastern Barents Sea, from reanalysis sea_ice_transport_across_line **SI_VT_BSE_Reanalysis [km^3 month^{-1}]** Sea ice volume transport across the eastern Barents Sea, from reanalysis sea_ice_transport_across_line **SI_AT_BSN_Reanalysis [km^2 month^{-1}]** Sea ice area transport across the northern Barents Sea, from reanalysis sea_ice_transport_across_line **SI_VT_BSN_Reanalysis [km^3 month^{-1}]** Sea ice volume transport across the northern Barents Sea, from reanalysis sea_ice_transport_across_line

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

An exhaustive description of the files nomenclature when downloaded through the web portal is available here : Files Nomenclature and Format

More information on product and dataset nomenclatures is available here Products and Datasets nomenclature

5) REFERENCES

Quality Information Document

https://documentation.marine.copernicus.eu/QUID/CMEMS-ARC-QUID-CLIMATE-SI-transport-NordicSeas.pdf
https://documentation.marine.copernicus.eu/QUID/CMEMS-ARC-QUID-002-003.pdf